

VINEET JETHANI

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EDUCATION

Sardar Vallabhbhai National Institute Of Technology
B.Tech in Electronics and Communication Engineering – 8.10 CGPA

Surat, Gujarat
Aug 2023 – Expected July 2027

Senior Secondary (CBSE)
Score – 93%

Surat, Gujarat
2023

Secondary (CBSE)
Score – 96.33%

Surat, Gujarat
2021

ACADEMIC PROJECTS

EEG-Based Seizure Prediction using Spiking Neural Networks

- Processed EEG signals (0.5–40 Hz) using window-based segmentation and Butterworth filtering for seizure prediction.
- Worked with labeled EEG datasets to identify ictal and non-ictal segments and performed statistical analysis on class distribution.
- Implemented a hybrid deep learning framework combining CNN-based feature extraction with an MLP classifier for EEG signal classification.
- Developed a spike-driven gating mechanism using LIF neuron to selectively process informative EEG windows, improving computational efficiency for real-time inference.
- Achieved **92.35%** accuracy, **92.38%** specificity, and **85.82%** sensitivity using LOPO-CV technique, and achieved an energy consumption of **151.53 μJ** per inference while deploying the model on NVIDIA Jetson Nano.

UART Transmitter

- Designed and implemented a UART Transmitter from scratch in Verilog with an internal baud rate generator derived from the system clock for accurate bit timing.
- Designed a 4-state FSM (IDLE, START, DATA, STOP) based control path ensuring protocol-correct 8N1 serial framing with LSB-first data serialization via a shift register datapath.
- Verified end-to-end functionality using a self-checking testbench in Xilinx Vivado; performed waveform-level timing analysis to validate setup/hold margins.

2G GSM Mobile Signal Jammer

- Designed a 900 MHz RF interference circuit using an NE555 astable multivibrator as the modulation source driving the base of a BF495 RF transistor configured in common-emitter topology.
- Designed an LC tank oscillator with air-core inductors and trimmer capacitors, achieving tunable output in the 880–960 MHz GSM band; calculated resonant frequency using $f = \frac{1}{2\pi\sqrt{LC}}$ and verified via LTspice simulation.
- Analyzed transistor biasing conditions (Q-point stability, V_{CE} , I_C) to ensure linear operation and maximize RF output power without entering saturation.

SKILLS

Area of Interest

Digital VLSI Design, FPGA Design, Digital Signal Processing

Core Subjects

Digital Electronics, CMOS VLSI Design, Computer Architecture

Simulation & Design Tools

Vivado, Cadence Virtuoso, LTspice, KiCAD

Hardware Description Languages

Verilog, VHDL

Programming

Python, C/C++ (basic)